Weather vs Energy Usage Prediction

Linear Regression Using Numerical Methods and Statistical Formulas

Subject: Numerical Methods

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Dataset: Appliances Energy Prediction Dataset

https://www.kaggle.com/code/gaganmaahi224/appliances-energy-time-series-analysis

Platform: Python (Google Colab / Jupyter Notebook)

GitHub Link: [Weather vs Energy Predection]

1. Introduction

Linear Regression is a fundamental statistical technique used to model the relationship between independent variables (X) and a dependent variable (Y). It is widely used for prediction and trend analysis.

In this project, linear regression is applied to predict **Appliances Energy Consumption** using weather-related factors. The implementation uses statistical analysis and regression formulas, along with Python libraries for computation and visualization.

The objective is to compute regression parameters, analyze data using statistical measures, evaluate prediction errors, and visualize results using professional plots.

The dataset used is the **Appliances Energy Prediction Dataset**.

2. Dataset Description

Dataset File:

energydata_complete.csv

Selected Features:

- T_out → Outside Temperature
- RH_out → Outside Humidity
- Windspeed → Wind Speed
- Appliances → Energy Consumption (Target Variable)

Regression Setup:

Inputs:

X1=T_out,

X2=RH_out,

X3=Windspeed

Output: $Y = Appliances$

Total Samples Used:

n=19,735

3. Data Cleaning and Preparation

The dataset was checked using:

- df.info()
- df.describe()
- df.isnull().sum()

No major missing values were found in the selected variables.

The data was split into:

- **Training Set:** 80%
- **Testing Set:** 20%

4. Statistical Calculations

The following statistical measures were calculated for each variable.

4.1 Appliances (Y)

- Mean = **97.6950**
- Median = **60.0000**
- Mode = **50.0000**
- Standard Deviation = **102.5223**

4.2 Outside Temperature (T_out)

- Mean = **7.4117**
- Median = **6.9167**
- Mode = **9.8000**

- Standard Deviation = **5.3173**

4.3 Outside Humidity (RH_out)

- Mean = **79.7504**
- Median = **83.6667**
- Mode = **95.0000**
- Standard Deviation = **14.9007**

4.4 Windspeed

- Mean = **4.0398**
- Median = **3.6667**
- Mode = **1.0000**
- Standard Deviation = **2.4512**

Formula Used

Mean: $\boldsymbol{\mu = \frac{1}{n}\sum X_i}$

Variance: $\sigma^2 = \frac{1}{n}\sum(X_i - \mu)^2$

Standard Deviation: $\boldsymbol{\sigma = \sqrt{\sigma^2}}$

5. Linear Regression Model

We used **Multiple Linear Regression** with three input variables.

General Equation: $Y = c + m_1X_1 + m_2X_2 + m_3X_3$

Where:

- c = Intercept

- m_1, m_2, m_3= Coefficients

Obtained Regression Coefficients

- Intercept (c) = **160.7888**
- Coefficient of T_out = **0.1754**
- Coefficient of RH_out = **-0.9374**
- Coefficient of Windspeed = **2.5651**

Final Regression Equation:

$$Appliances = 160.79 + 0.18(T_{out}) - 0.94(RH_{out}) + 2.57(Windspeed)$$

6. Prediction and Residuals

Predicted values were generated using the regression model.

Residual Formula:

$$e_i = Y_i - \widehat{Y_i}$$

Where:

Y_i= Actual Value

$\widehat{Y_i}$= Predicted Value

Residuals were used to analyze model errors.

7. Error Metrics

The following metrics were used to evaluate model performance.

Mean Absolute Error:

$$\text{MAE}=\text{n}1\sum| e_i|$$

Root Mean Squared Error:

$$RMSE = \sqrt{\frac{1}{n}\sum e_i^2}$$

Coefficient of Determination:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Metric Values

- **MAE = 59.0016**
- **RMSE = 101.1292**
- **R^2 = 0.026991**

Accuracy:

0.026991×100=2.70%

8. Graphical Analysis

The following plots were generated:

1. Scatter Plot + Regression Line
2. Sorted Actual vs Predicted Plot
3. Residual Plot
4. Histogram of Residuals

Output File:

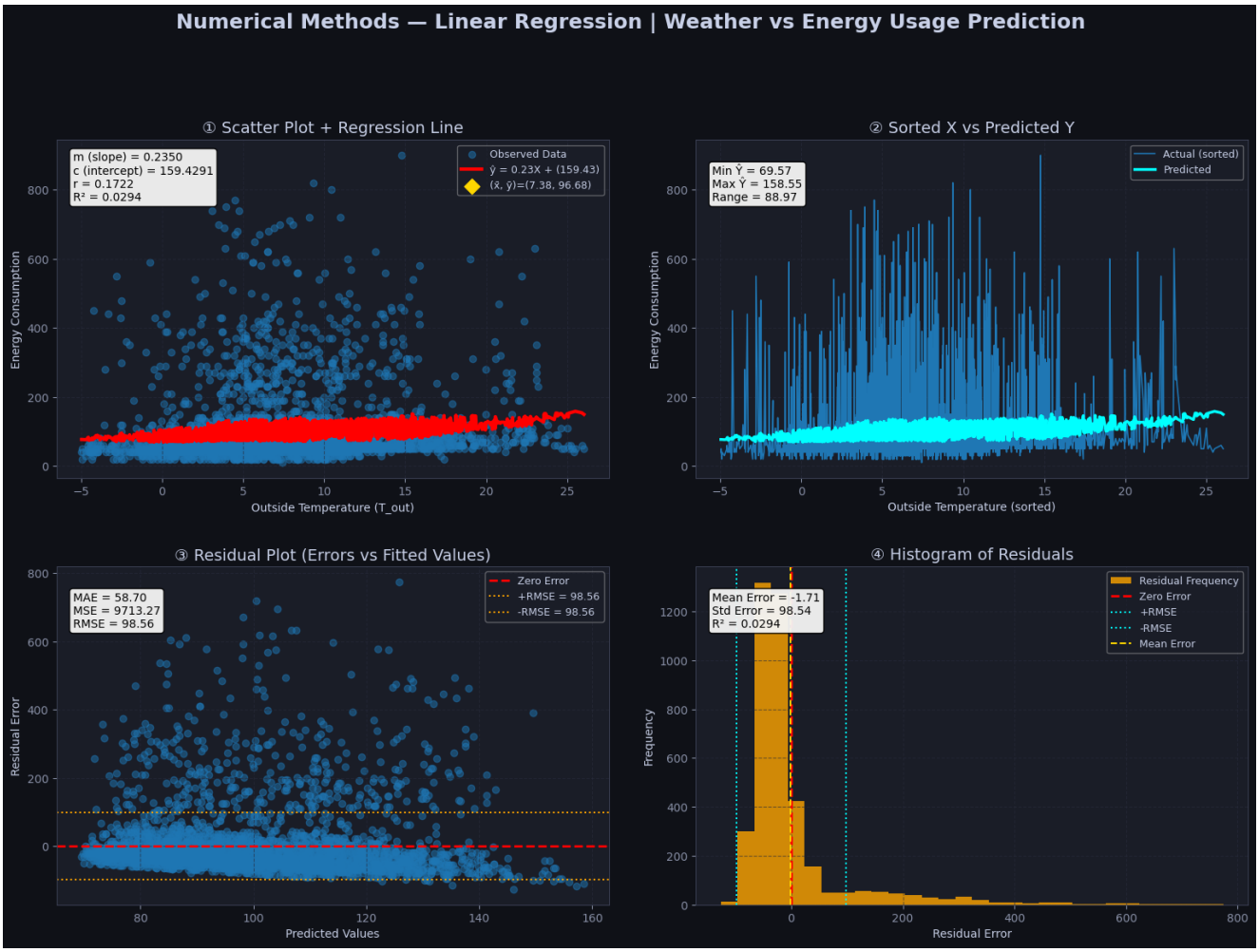

Figure 1: Final Regression Dashboard

9. Results Summary Table

Category	Metric	Value
Dataset	Total Samples	19,735
Variables	Input X	T_out, RH_out, Windspeed
Variables	Output Y	Appliances
Central Tendency (Y)	Mean	97.6950
Central Tendency (Y)	Median	60.0000

Central Tendency (Y)	Mode	50.0000
Dispersion (Y)	Std. Deviation	102.5223
Regression Coefficients	Intercept	160.7888
Regression Coefficients	T_out	0.1754
Regression Coefficients	RH_out	-0.9374
Regression Coefficients	Windspeed	2.5651
Error Metrics	MAE	59.0016
Error Metrics	RMSE	101.1292
Model Accuracy	R^2	0.026991 (2.70%)
Output	Plots File	weather_energy_dashboard.png

10. Observations

- Outside temperature has a small positive effect on energy consumption.
- Outside humidity negatively affects energy usage.
- Windspeed shows positive influence.
- Model accuracy is low because appliance energy consumption also depends on occupancy, time of day, internal room temperature, and human behavior.
- Weather variables alone are insufficient for high-accuracy prediction.

11. Conclusion

This project successfully demonstrates the implementation of **Linear Regression** using statistical analysis and Python tools.

All major concepts such as:

- Mean, Median, Mode
- Standard Deviation
- Regression Coefficients

- Prediction
- Residual Analysis
- Error Metrics
- Visualization

were successfully applied.

The project validates how regression works in real-world datasets and provides strong practical understanding of Numerical Methods.

12. Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab